

Set notation



1

a $S = \{1, 2, 3, 4, 5, 6\}$

b $S = \{1, 3, 5, 7, 9\}$

2 $\{5, 6, 7\}$

3

a People who do not like softball.

b People who like either football, softball or swimming.

4

a Yes

b No

Intersection and union

5

a The set of all the elements in the universal set that are not in set A .

b The set containing all the elements that are members of set A or of set B or of both sets.

c The set containing all the elements that are common to both set A and set B .

6

a $\{1, 2, 4, 8, 16\}$

b $\{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 15, 16, 17, 21\}$

7 $\{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$

8

a $A \cap B = \{0, 4, 16, 36, 64\}$

b $A \cap B = \{5, 10, 15, 20, 25, 30, 35, 40, 45\}$

c $A \cap B = \{3, 7\}$

9

a False

b False

10

a True

b True

11



a

i {1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36}

ii {1, 2, 3, 4, 6, 12}

b

i {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 17, 19, 23, 29, 31, 37}

ii {2, 3, 5, 7, 11}

12

a {1, 2, 3, 4, 5, 6}

b {1, 2, 3, 4, 5, 6, 7, 8}

c {1, 2, 3, 4, 5, 6}

d {1, 2, 3, 4, 5, 6, 7, 8}

13

a

i The universal set.

ii The universal set, except for the elements of A and the elements of B .

b

i The universal set.

ii The universal set, except for the elements of A and the elements of B .

14

a The universal set, except for the elements that are in both A and B .

b The set of common multiples of 4 and 6 greater than or equal to 70.

c No

15

a The set of integers that are not multiples of 4 less than 70 or multiples of 6 less than 70.

b The set of integers that are not less than 70, and are multiples of both 4 and 6.

c No

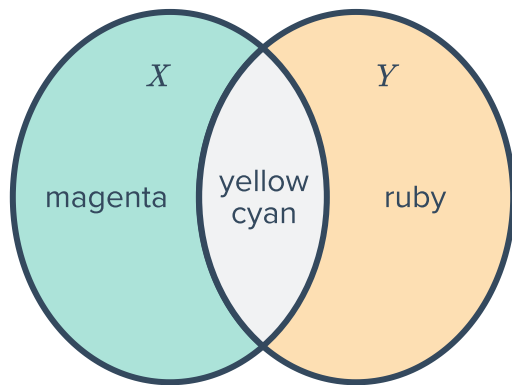
16 Yes



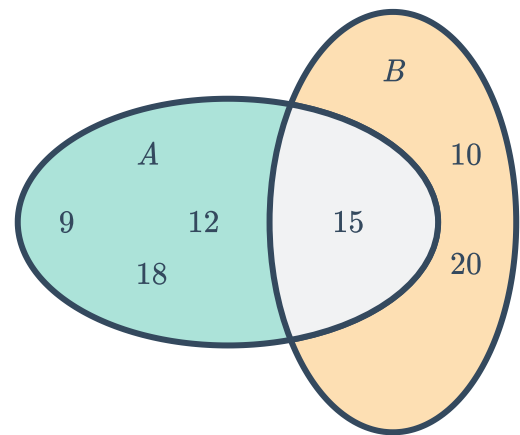
Venn diagrams

17

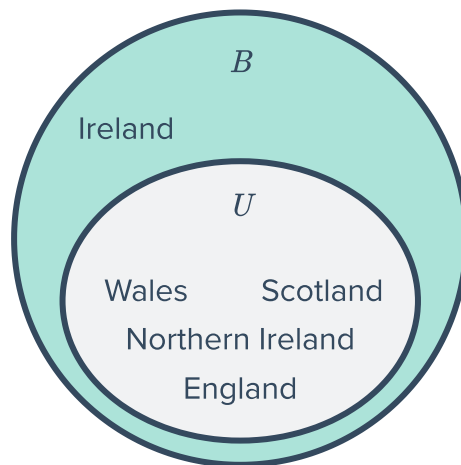
a



b



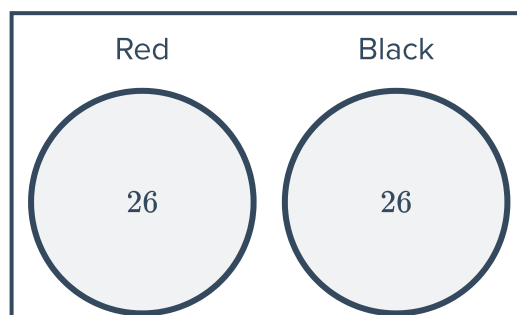
c



18 No

19 4 regions

20





21

a $A' = \{7, 14, 12, 10\}$

b $B' = \{4, 5, 1, 12, 10\}$

22

a $A \cap B = \{7, 8\}$

b $A \cup B = \{2, 4, 6, 7, 8, 10, 14\}$

23

a $A = \{1, 3, 4, 5, 9, 10\}$

b $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

c $B' = \{1, 2, 9, 10, 11, 13\}$

24

a $A \cap B' = \{3, 4, 6\}$

b $(A \cup B)' = \{15, 60\}$

25

a $A \cap C = \{4, 5, 9\}$

b $(B \cap C)' = \{1, 2, 3, 7, 9, 10, 11, 12, 13, 14\}$

c $A \cap B \cap C = \{4, 5\}$

26

a Yes

b No

27 No

28

a $A \cap (B \cup C) = \{3, 4, 5, 9\}$

b $(A \cap B)' = \{1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

29

a 2

b 5

c 3

d 4

e 8

f 7

g 1

h 6

30

a Yes

b Yes

c The elements d and e are in both sets, so if we try to find the total number of elements in either set by adding $n(A)$ and $n(B)$, we will be counting these elements twice. Therefore, we will have to subtract the number of elements common to both sets.

31 54



32 48

33

- a Venn diagram 4
- c Venn diagram 2

- b Venn diagram 3
- d Venn diagram 1

34

- a 169
- b 223
- c 192
- d 138
- e 262
- f 293

35

	Play Rugby League	Don't play Rugby League
Play Rugby Union	43	185
Don't play Rugby Union	179	146

Mutually exclusive events

36 C, D

37

- a
 - i $x = 0$
 - ii Mutually exclusive
- b
 - i $x = 0$
 - ii Mutually exclusive

38

a

i 0.3

iii 0.1

b

i 0.2

iii 0

c

i 0.6

iii 0.2

d

i 0.1

iii 0



ii 0.2

iv Non-mutually exclusive

ii 0.1

iv Mutually exclusive

ii 0.1

iv Mutually exclusive

ii 0.7

iv Mutually exclusive

39

a 0

b Yes

40 0.36

41 $\frac{3}{4}$

Non-mutually exclusive events

42 The probability of getting a 1, 3, 5 or 6.

43 $P(A \cup B) = \frac{1}{3}$

44 $P(A \cap B) = \frac{2}{5}$

45 $P(B) = \frac{1}{5}$

46 $P(A \text{ or } B) = 0.95$